SEC P vs NP log 2026-05-09

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-13 12:30 UTC

SEC P vs NP — daily log 2026-05-09

70 cycles recorded

Submodular Width Separates Monotone DNF from k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Submodular Function Optimization × Monotone DNF Formulas for k-CLIQUE
- **Recorded**: 2026-05-09 00:12 UTC
- **Entry ID**: `887c2309e24e`

Statement

For any monotone DNF formula F on n variables, the submodular width μ(F) is O(log n), but for the k-CLIQUE indicator function, μ is Ω(n).

Rationale

Submodular functions capture the diminishing returns property, which might reflect the structure of DNF formulas. The k-CLIQUE problem's inherent complexity could lead to higher submodular width, providing a lower bound.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.05s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

---

Noncommutative Fourier Coefficient Norm Separates Read-Twice from Read-Once BPs

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Noncommutative Harmonic Analysis × Communication Complexity Lower Bounds for Read-Twice Branching Programs
- **Recorded**: 2026-05-09 00:21 UTC
- **Entry ID**: `3dbeb5b4ba3d`

Statement

For any read-once BP P, the norm of its noncommutative Fourier coefficients is O(log n), while for any read-twice BP P, it is Ω(n). This separation implies that distinguishing between read-once and read-twice BPs requires communication complexity Ω(n).

Rationale

Noncommutative harmonic analysis captures the non-abelian structure of read-twice BPs via group representations, which read-once BPs lack. The exponential growth of the norm for read-twice BPs reflects their inherent complexity, enabling Fourier-analytic lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-9`, elapsed: `82.37s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Re-run test with detailed logging to identify crash cause and collect empirical results

---

Symmetric Group Fourier Min-Coefficient Lower Bound on Disjointness

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Transform on Symmetric Group × Communication Complexity of Disjointness
- **Recorded**: 2026-05-09 00:31 UTC
- **Entry ID**: `96f8e73f18cc`

Statement

For any n, the minimal Fourier coefficient of the communication matrix M for the disjointness problem is Ω(n), which lower bounds the randomized communication complexity of the problem.

Rationale

The Fourier transform on the symmetric group captures the combinatorial structure of disjointness matrices via permutation symmetries. The minimal coefficient corresponds to the 'most uniform' contribution of permutation orbits, which inherently requires Ω(n) communication to distinguish.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 1.74s


```
tances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
TRIAL: {'metric_name': 'min_fourier_coefficient', 'metric_value': 40.0, 'instances_tested': 30, 'con
RESULT: FALSIFIED counterexample="" first_failing_seed=None
```


Judge reasoning

Safety rail: critic_challenge | original judge: All 30 trials confirmed the conjecture holds with min Fourier coefficient 40.0, meeting the support fraction threshold. | next: Test larger n values to verify scalability of the lower bound

---

Plethysm Coefficient Inverse Proportionality to Circuit Size

- **Verdict**: INCONCLUSIVE
- **Bridge**: Plethysm Theory × Boolean Circuit Size
- **Recorded**: 2026-05-09 00:45 UTC
- **Entry ID**: 9e8836004e56

Statement

For every CNF formula Φ with n variables, the plethysm coefficient of its associated symmetric function f_Φ(λ) satisfies plethysm(f_Φ, λ) ≤ 2^{O(n)} / size(Φ) for all partitions λ ⊢ n.

Rationale

Plethysm coefficients encode representation-theoretic multiplicities that may reflect combinatorial constraints on function decomposition. By mapping CNF clauses to symmetric functions, we expose potential algebraic barriers to efficient circuit synthesis.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s


```
ue, 'counterexample': ''}
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
```

```
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0
```

Judge reasoning

Safety rail: critic_challenge | original judge: All 12 trials confirmed the conjecture holds with metric_value=0, meeting the 80% support threshold. No counterexamples found. | next: Test conjecture on non-trivial CNF formulas with varying circuit sizes to validate across diverse cases

---

Free Entropy Inverse Proportionality to Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 00:51 UTC
- **Entry ID**: `e9bfefbfd0c1`

Statement

For every n ≥ 1, the free entropy Φ(M_n) of the communication matrix M_n for the DISJOINTNESS problem satisfies Φ(M_n) ≥ c / (n log n), where c > 0 is a universal constant. Equality holds if and only if M_n is a tensor product of n-dimensional projections.

Rationale

Free entropy quantifies the non-commutative randomness of a matrix. DISJOINTNESS's Ω(n) communication lower bound suggests strong non-commutative structure. Free entropy's inverse scaling with n mirrors the information-theoretic cost of coordination in distributed computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e19bcb97.py", line 115, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e19bcb97.py", line 95, in run_trial
    eigs = eigenvalues(X)
           ^^^^^^^^^^^^^^
```

```
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e19bcb97.py", line 71, in eigenvalues
   h = sum(A[i][i+k-1]**2 for i in range(k-1))
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e19bcb97.py", line 71, in <genexpr>
   h = sum(A[i][i+k-1]**2 for i in range(k-1))
           ~~~~^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError, preventing data collection to evaluate conjecture | next: Debug the index error in eigenvalues calculation by adding boundary checks for matrix dimensions

---

Young Tableaux Count Lower Bound on Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Communication Complexity (Disjointness Problem)
- **Recorded**: 2026-05-09 00:57 UTC
- **Entry ID**: `221ada155c09`

Statement

For a disjointness problem with n variables, the communication complexity is at least Ω(√(number of standard Young tableaux of shape (n, n))), where the number is calculated via the hook-length formula.

Rationale

The number of standard Young tableaux of shape (n, n) corresponds to the dimension of the symmetric group representation, which might influence the rank of the communication matrix, leading to a lower bound on communication complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cfad8ba1.py", line 76, in <module>
   result = run_trial(seed)
            ^^^^^^^^^^^^^^^
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cfad8ba1.py", line 48, in run_trial
   tableaux_count = hook_length_formula(n)
                    ^^^^^^^^^^^^^^^^^^^^^^
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cfad8ba1.py", line 24, in hook_length_f
   return math.factorial(n**2) // det
          ~~~~~~~~~~~~~~~~~~~~~^^~~~~
ZeroDivisionError: integer division or modulo by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of the conjecture's validity | next: Fix the hook-length formula implementation to handle division by zero and re-run tests

---

Symmetric Polynomial Monomial Count and Resolution Proof Size for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symmetric Polynomials (Algebraic Combinatorics) × Resolution Proof Size (Tseitin Formulas)
- **Recorded**: 2026-05-09 01:10 UTC
- **Entry ID**: `5421dda86f76`

Statement

For a Tseitin formula derived from a bipartite graph with n edges, the size of the minimal resolution proof is Θ(m), where m is the number of distinct monomials in the symmetric polynomial generated by the graph's adjacency matrix.

Rationale

The symmetric polynomial's monomial count reflects the graph's structural complexity, which may impose constraints on the resolution steps required to derive a contradiction, leading to a proportional increase in proof size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
ome/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  [Previous line repeated 27 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 124, in dfs
    dfs(j, path)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  [Previous line repeated 27 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 124, in dfs
    dfs(j, path)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
```

```
    dfs(node + 1, path + [node])
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  [Previous line repeated 27 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 124, in dfs
    dfs(j, path)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1fee859.py", line 120, in dfs
    dfs(node + 1, path + [node])
  [Previous line repeated 27 more times]
  File "/home/ludo/
```

Judge reasoning

Test crashed during execution, preventing data collection to verify conjecture | next: Debug the recursive DFS implementation to identify stack overflow cause

---

Valuation Rank of Polynomial Coefficients Bounds ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Valuation Theory × ACC^0 Circuit Size
- **Recorded**: 2026-05-09 01:41 UTC
- **Entry ID**: `593ecfc3498f`

Statement

For any CNF formula Φ with n variables, the number of distinct p-adic valuations of its clause-encoding polynomial's coefficients is Θ(log n) iff Φ has an ACC^0 circuit of size O(n^3).

Rationale

Valuations capture arithmetic structure that may correlate with circuit complexity. The p-adic valuation's rank could expose algebraic constraints on polynomial representations of Boolean functions, linking valuation theory to circuit size bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe5b1309.py", line 93, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe5b1309.py", line 58, in run_trial
    valuations = [p_adic_valuation(coeff, p) for coeff in poly.coefficients()]
                                                           ^^^^^^^^^^^^^^^^^

AttributeError: 'int' object has no attribute 'coefficients'
```

Judge reasoning

Test crashed due to AttributeError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the test code to handle integer coefficients properly and re-run experiments with multiple seeds

---

Plethysm Coefficient Inverse Proportionality to SOS Refutation Size for Symmetric CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Plethysm Theory × SOS Refutation Size for CSPs
- **Recorded**: 2026-05-09 02:24 UTC
- **Entry ID**: `76c4f54f8fcf`

Statement

For a symmetric Boolean CSP instance Φ with n variables, the plethysm coefficient χ(Φ) of its characteristic polynomial satisfies χ(Φ) ≥ c · (log n) / (SOS_refutation_size(Φ)), where c > 0 is a constant. Equality holds when Φ is a symmetric constraint satisfaction problem with uniform clause distribution.

Rationale

Plethysm coefficients capture the complexity of decomposing symmetric representations, while SOS refutation size measures the hierarchy's ability to handle symmetric constraints. Their inverse proportionality suggests structural barriers where symmetric algebraic invariants limit SOS efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.17s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_18142075.py", line 58, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_18142075.py", line 35, in run_trial
    chi = plethysm_coefficient(n)
          ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_18142075.py", line 26, in plethysm_coef
    coeff += (-1) ** (n - k) * Fraction(math.comb(n, k)) * plethysm_coefficient(k)
                                                           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_18142075.py", line 26, in plethysm_coef
    coeff += (-1) ** (n - k) * Fraction(math.comb(n, k)) * plethysm_coefficient(k)
                                                           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_18142075.py", line 26, in plethysm_coef
    coeff += (-1) ** (n - k) * Fraction(math.comb(n, k)) * plethysm_coefficient(k)
                                                           ^^^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 993 more times]
  File "/usr/lib/python3.12/fractions.py", line 630, in reverse
    return monomorphic_operator(Fraction(a), b)
```

```
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 754, in _mul
    return Fraction._from_coprime_ints(na * nb, db * da)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

Test crashed with recursion depth error, preventing evaluation of conjecture validity | next: Implement memoization for plethysm_coefficient recursion or increase recursion limit

---

Young Tableaux Ratio Exponential Lower Bound on Permanent Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Boolean Circuit Size
- **Recorded**: 2026-05-09 02:52 UTC
- **Entry ID**: `37932008c65e`

Statement

For random n×n matrices, the ratio of standard Young tableaux counts for shape (n,n,...,n) (n times) to shape (n,n-1,...,1) is Ω(2^n). This implies permanent requires Ω(2^n) size circuits for n×n matrices.

Rationale

Young tableaux counts encode symmetry-breaking patterns in tensor decompositions. The exponential gap in tableau ratios reflects the inherent asymmetry between permanent's fully symmetric structure and determinant's staircase symmetry, suggesting computational hardness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.84s`

```
ith n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 32768, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 32768, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1099511627776, 'instances_tested': 0,
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1073741824, 'instances_tested': 0, 'co
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1024, 'instances_tested': 0, 'conjectu
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1073741824, 'instances_tested': 0, 'co
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1024, 'instances_tested': 0, 'conjectu
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1024, 'instances_tested': 0, 'conjectu

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_90d1a1b1.py", line 87, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_90d1a1b1.py", line 87, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
```

```
                                          ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

parse_fail | next: NONE

Plethysm Coefficient Exponential Gap in Symmetric Squares of Permutation Representations for Random CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Plethysm Theory × Boolean Circuit Size
- **Recorded**: 2026-05-09 03:07 UTC
- **Entry ID**: `c93f42c3f6f7`

Statement

For a random 3-CNF formula Φ with n variables and m clauses, let π(Φ) denote the plethysm coefficient of its characteristic polynomial under the symmetric square representation of S_n. Then π(Φ) ≥ 2^{n/2} · poly(m) with probability ≥ 2/3, while the determinant's symmetric square plethysm coefficient is bounded by poly(n,m).

Rationale

Plethysm coefficients capture the multiplicity of irreducible representations in symmetric power decompositions. The permanent's higher multiplicities in symmetric squares may reflect its inherent complexity, while the determinant's bounded coefficients suggest structural simplicity. This gap could imply circuit size separations via representation-theoretic obstructions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_258cdfd5.py", line 105, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_258cdfd5.py", line 84, in run_trial
    plethysm = plethysm_coefficient(char_poly, n)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_258cdfd5.py", line 63, in plethysm_coef
    return sum(hook_length_formula(shape, n) * char_poly[shape] for shape in partitions)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_258cdfd5.py", line 63, in <genexpr>
    return sum(hook_length_formula(shape, n) * char_poly[shape] for shape in partitions)
                                               ~~~~~~~~~^^^^^^^
TypeError: unhashable type: 'list'
```

Judge reasoning

Test crashed due to unhashable type error, preventing data collection. Cannot assess conjecture validity without successful runs. | next: Fix the char_poly data structure to use a dictionary instead of list for hashable lookups

---

Schur-Weyl Multiplicity Exponential Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Size
- **Recorded**: 2026-05-09 03:18 UTC
- **Entry ID**: `7da975152a0e`

Statement

For a random 3-CNF formula Φ on n variables, let m_Φ(λ) denote the multiplicity of irreducible representation λ in the decomposition of Sym^k(V) ⊗ Sym^k(V) under the action of GL_n(C), where V is the permutation representation of S_n. Then m_Φ(λ) ≥ 2^{Ω(n)} for λ = (n−1,1) iff Φ is a permanent-like formula, while m_Φ(λ) ≤ 2^{O(√n log n)} for λ = (n−1,1) iff Φ is a determinant-like formula.

Rationale

Schur-Weyl multiplicities encode symmetry structures of tensor powers, which may reflect inherent complexity gaps between permanent and determinant. Exponential vs subpolynomial growth in specific Young diagrams could mirror circuit size separations via representation-theoretic obstruction theory.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67a15ccc.py", line 118, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67a15ccc.py", line 91, in run_trial
    permanent_multiplicity = schur_weyl_multiplicity(n, k, λ)
                             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67a15ccc.py", line 83, in schur_weyl_mu
   return sum(schur_weyl_multiplicity_for_partition(partition) for partition in combinations(range(k
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67a15ccc.py", line 83, in <genexpr>
   return sum(schur_weyl_multiplicity_for_partition(partition) for partition in combinations(range(k
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67a15ccc.py", line 80, in schur_weyl_mu
    multiplicity += schur_weyl_coefficient(λ, μ)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67a15ccc.py", line 73, in schur_weyl_co
```

```
       product  *=  hook_length_formula(n,  λ[i])  //  (hook_length_formula(n,  λ[i]  -
μ[i]) * hook_length_formula(n, μ[i]))
         ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
ZeroDivisionError: integer division or modulo by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of conjecture validity | next: Fix division-by-zero in hook_length_formula implementation and re-run tests

---

SOS Refutation Degree and Convex Body Volume for Random 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Convex Geometry × SOS Refutation Degree for 3-SAT Instances
- **Recorded**: 2026-05-09 03:44 UTC
- **Entry ID**: `ebd764aaaa68`

Statement

For a random 3-SAT instance with n variables, the volume of the feasible region in the SOS relaxation of degree d is exponentially smaller than the volume of the entire space if and only if the minimal refutation degree required to refute the instance is $\Omega(\log n)$.

Rationale

The SOS hierarchy's relaxations can be viewed as convex bodies, and their volumes can be analyzed using convex geometry. The exponential reduction in volume could indicate a structural property that allows for efficient refutations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
n degree not Ω(log n)'}
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
TRIAL: {'metric_name': 'Volume Ratio', 'metric_value': 3.863855686442174e-30, 'instances_tested': 1,
RESULT: FALSIFIED counterexample="minimal refutation degree not Ω(log n)" first_failing_seed=11
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: The counterexample shows the volume ratio is exponentially small even when the minimal refutation degree is not $\Omega(\log n)$, directly contradicting the c | next: Analyze the specific counterexample instance to identify structural properties that invalidate the conjecture's equivalence

---

Random Matrix Eigenvalue Deficit and SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Degree for Max-CUT
- **Recorded**: 2026-05-09 03:53 UTC
- **Entry ID**: `7eb129929a17`

Statement

For a random Max-CUT instance with n variables, the number of eigenvalues of the adjacency matrix lying in $[-\lambda, \lambda]$ is $\Theta(\sqrt{n})$, where $\lambda = 1/\sqrt{n}$. The SOS degree required to approximate Max-CUT with 0.878-accuracy is at least the count of eigenvalues outside $[-\lambda, \lambda]$.

Rationale

The spectral properties of the adjacency matrix encode structural constraints on the Max-CUT problem. Eigenvalues outside $[-\lambda, \lambda]$ indicate non-trivial correlations that necessitate higher SOS degrees to capture the problem's geometry, as per the Goemans-Williamson framework.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5df4622f.py", line 83, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5df4622f.py", line 64, in run_trial
    eigvals = eigenvalues(A)
              ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5df4622f.py", line 52, in eigenvalues
    eigval = power_method(A)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5df4622f.py", line 38, in power_method
    x /= math.sqrt(sum(x[i]**2 for i in range(n)))
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

Test crashed during eigenvalue computation, preventing data collection | next: Fix the power_method division error and revalidate numerical stability

---

Real Radical Rank Bounds SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-09 04:05 UTC
- **Entry ID**: `f54522e2e780`

Statement

For a random Max-CUT instance on n variables, the minimal SOS degree d(n) required to approximate the maximum cut satisfies d(n) ≥ log2(rank(ℝ-radical(I))), where I is the ideal of polynomials vanishing on the feasible region of the instance.

Rationale

The real radical of an ideal captures the semialgebraic structure of the feasible region. Its rank, as a measure of algebraic complexity, may correlate with the SOS hierarchy's depth needed to certify optimality, offering a geometric lens on convex relaxation trade-offs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6b41a151.py", line 118, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6b41a151.py", line 101, in run_trial
    rank_I = real_radical_rank([instance])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6b41a151.py", line 76, in real_radical_
    A = [[I[i][j] * I[k][l] for l in range(n)] for k in range(n) for j in range(n)]
            ^
NameError: name 'i' is not defined. Did you mean: 'I'?
```

Judge reasoning

Test crashed due to NameError, preventing data collection to evaluate conjecture | next: Fix the NameError in real_radical_rank function by defining variables properly

---

Discriminant Exponentiation Bounds SOS Refutation Degree for Random 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Discriminant Theory) × SOS Refutation Degree for 3-SAT Instances
- **Recorded**: 2026-05-09 04:26 UTC
- **Entry ID**: `149197523066`

Statement

For a random 3-SAT instance with n variables, the discriminant of its polynomial system (encoding clauses as degree-2 polynomials) satisfies disc(Φ) ≥ 2^{Ω(n)} if and only if the SOS refutation degree for Φ is Ω(√n).

Rationale

Discriminants capture the geometric complexity of solution sets; their exponential growth correlates with the SOS hierarchy's need for deeper certificates to certify infeasibility in highly constrained systems.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.16s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_93eae6dd.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_93eae6dd.py", line 117, in run_trial
    disc = discriminant(poly)
           ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_93eae6dd.py", line 58, in discriminant
    A[int(var) - 1][int(var) - 1] += coeff ** 2
      ^^^^^^^^
ValueError: invalid literal for int() with base 10: '1*xx1'
```

Judge reasoning

Test crashed due to invalid literal conversion, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug discriminant function's variable parsing logic to handle clause encodings properly

---

Spectral Gap Exponential Lower Bound on Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Proof Length
- **Recorded**: 2026-05-09 04:34 UTC
- **Entry ID**: `cd7a96f56994`

Statement

For a d-regular graph G with second eigenvalue λ_2, the Tseitin formula on G requires Resolution refutations of length ≥ 2^{Ω(1/λ_2)}. For graphs with λ_2 ≥ 1 - ε, the refutation length is polynomial in n.

Rationale

Expander graphs (small λ_2) force long Resolution proofs via cumulative entropy arguments. The spectral gap directly quantifies expansion properties, linking graph structure to proof complexity via the interplay between clause dependencies and eigenvalue decay.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
TRIAL: {'metric_name': 'Tseitin Formula Length', 'metric_value': 32.0, 'instances_tested': 1, 'conjec

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6219b7f4.py", line 79, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6219b7f4.py", line 60, in run_trial
    G = generate_d_regular_graph(n, d)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6219b7f4.py", line 31, in generate_d_re
    raise ValueError("d must be even")
ValueError: d must be even
```

Judge reasoning

Test crashed due to invalid d parameter (must be even), preventing valid data collection. Pre-registered support condition (80% seeds) cannot be evaluated. | next: Fix test parameters to use even d values for d-regular graph generation

---

Dehn Function Exponentiation Bounds Resolution Length for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Group Theory × Resolution Proof Length
- **Recorded**: 2026-05-09 04:46 UTC
- **Entry ID**: `4d96b3fd6533`

Statement

For a Tseitin formula $\Phi(G)$ derived from a d-regular graph G on n vertices, let $\delta(G)$ denote the minimal Dehn function of a group presentation constructed from G's adjacency matrix. Then resolution length($\Phi(G)$) ≥ 2^{$\Omega(\delta(G))$}. For non-expanders, $\delta(G) \leq$ poly(n) and resolution length($\Phi(G)$) = poly(n).

Rationale

Dehn functions measure the complexity of filling loops in group presentations, mirroring the combinatorial complexity of resolution proofs. By linking group-theoretic filling properties to proof complexity, we expose structural parallels between geometric group theory and proof systems. Expanders, with their high Dehn functions, would require exponentially longer proofs, while non-expanders have polynomial bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cdb7beb6.py", line 120, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cdb7beb6.py", line 92, in run_trial
    delta_G = dehn_function(G)
              ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cdb7beb6.py", line 53, in dehn_function
    A = gaussian_elimination(G + I)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cdb7beb6.py", line 43, in gaussian_elim
    A[i][j] *= factor
    ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during Gaussian elimination with index error, preventing data collection. Pre-registered support condition cannot be evaluated. **|** next: Fix matrix indexing in Gaussian elimination and revalidate dehn_function implementation

---

Symmetric Power Rank Gap in Permanent vs Determinant Polynomials

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Symmetric Power Rank of Homogeneous Polynomials
- **Recorded**: 2026-05-09 05:35 UTC
- **Entry ID**: `d27b04e7091e`

Statement

For all n ≥ 2, the symmetric power rank of the permanent polynomial Perm_n is at least 2^{n/2} times the symmetric power rank of the determinant polynomial Det_n for exponents k = ⌈n/2⌉.

Rationale

Symmetric power ranks capture the complexity of decomposing polynomials into symmetric tensors, which mirrors the algebraic structure of permanent/determinant. The exponential gap conjectured here would imply inherent structural differences in their representation-theoretic decompositions, potentially separating their complexity classes.

Novelty

- Judge: `NOVEL` over 5 arXiv hits

Top hits consulted: - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent - [1101.2456v3] Polynomial representations and categorifications of Fock Space - [1603.02493v4] Clifford theory for glider representations - [1604.00993v1] The 'Core' of Symmetric Homogeneous Polynomial Inequalities of Degree Four of Three Real Variables - [2206.05177v2] m-Symmetric functions, non-symmetric Macdonald polynomials and positivity conjectures

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9ba6ff81.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9ba6ff81.py", line 66, in run_trial
    threshold = Fraction(2**(n/2), 10)
                ^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances
```

Judge reasoning

Test crashed due to invalid Fraction arguments; no data produced to assess conjecture. | next: Fix the Fraction constructor to use integer exponents (e.g., 2**(n//2)) and re-run tests

---

Real Radical Dimension Inverse Proportionality to SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-09 05:41 UTC
- **Entry ID**: `fcdff37e3de9`

Statement

For any Max-CUT instance with n variables, the minimal SOS degree d required to achieve a 0.879-approximation satisfies d ≥ dim(ℝ-rad(I)), where I is the ideal generated by the Max-CUT constraints and ℝ-rad(I) is its real radical.

Rationale

The real radical captures the essential structure of the constraints, and its dimension reflects the complexity needed to represent the feasible solutions. A lower SOS degree would insufficiently capture this structure, leading to approximation errors.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
) = 7 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 6 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 11 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 10 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 15 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 8 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 13 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 10 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 12 > d_SOS = 2'}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Fa
rad(I)) = 13 > d_SOS = 2'}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1ab17c32.py", line 132, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1ab17c32.py", line 132, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed with KeyError 'seed' before completing evaluation; pre-registered support condition cannot be verified. | next: Fix the test code to handle missing 'seed' keys and re-run with proper error handling

---

Free Entropy Lower Bound on Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 06:02 UTC
- **Entry ID**: `54cd565a7ba3`

Statement

For a random instance M of the DISJOINTNESS problem on n-bit inputs, the free entropy $\phi(M)$ satisfies $\phi(M) \geq c \cdot n$ for some absolute constant $c > 0$. This lower bound is tight up to polynomial factors.

Rationale

Free entropy captures noncommutative information structure, which may mirror the entanglement required for disjointness protocols. The conjecture links operator-algebraic invariants to communication complexity via matrix-valued distributions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8acca6bd.py", line 58, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8acca6bd.py", line 30, in run_trial
    ϕ_M = r_transform(M) / (2 * math.pi)
          ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8acca6bd.py", line 28, in r_transform
    return math.log(abs(det))
           ^^^^^^^^^^^^^^^^^^
ValueError: math domain error
```

Judge reasoning

Test crashed with math domain error, preventing evaluation of ϕ(M). The pre-registered support condition cannot be verified without successful trials. | next: Debug the r_transform function's determinant calculation to avoid math domain errors, particularly when det approaches zero

---

Fourier Coefficient Decay and Resolution Proof Length for k-CNF Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis on Boolean Functions × Resolution Proof Length
- **Recorded**: 2026-05-09 06:15 UTC
- **Entry ID**: `3a56915c32e1`

Statement

For a k-CNF formula Φ with n variables, the resolution proof length is Θ(2^{n/2} / μ), where μ is the maximum absolute value of Φ's Fourier coefficients. For all Φ with μ ≤ 1/2^{n/2 - 5}, the proof length exceeds 2^{n/2 - 5} × n^2.

Rationale

The Fourier coefficients capture variable influence; exponential decay suggests limited variable interaction, implying shorter proofs. The inverse proportionality links structural simplicity (small μ) to proof complexity, with a polynomial factor accounting for clause density.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34334eed.py", line 110, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34334eed.py", line 81, in run_trial
    fourier_coeffs = [0] * (1 << n)
                     ~~~~^~~~~~~~~~
MemoryError
```

Judge reasoning

Test crashed with MemoryError, preventing evaluation of conjecture validity | next: Optimize memory usage for large n or implement incremental Fourier coefficient calculation

---

Hilbert Function Leading Coefficient Distinguishes Read-Twice from Read-Once BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry × Read-Twice vs Read-Once Branching Programs
- **Recorded**: 2026-05-09 06:21 UTC
- **Entry ID**: `4febd18902d4`

Statement

For a read-twice BP P over n variables, the leading coefficient of the Hilbert function of the ideal I_P generated by its computation paths satisfies deg(I_P) = Ω(n), while for read-once BPs, deg(I_P) = O(log n).

Rationale

The Hilbert function's growth rate reflects the dimension of the ideal's variety, which encodes the BP's computational structure. Read-twice BPs generate higher-dimensional ideals due to repeated variable access, creating a detectable algebraic signature.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96cab250.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96cab250.py", line 60, in run_trial
    read_twice_deg = hilbert_function(read_twice_monomials)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96cab250.py", line 45, in hilbert_funct
    degree = sum(monomial)
             ^^^^^^^^^^^^^
TypeError: unsupported operand type(s) for +: 'int' and 'str'
```

Judge reasoning

Test crashed due to type error in monomial summation, preventing data collection | next: Debug monomial processing in hilbert_function to handle type conversions properly

---

Moment Matrix Rank Inverse Proportional to SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-09 06:33 UTC
- **Entry ID**: `64023e1d5813`

Statement

For every Max-CUT instance on n variables, the minimal SOS degree d required to achieve an approximation ratio of 0.878 + ε satisfies d ≥ log(rank(M)), where M is the moment matrix of the SOS relaxation. This holds for all instances with spectral norm ||A|| ≤ 1 + 1/n.

Rationale

The rank of the moment matrix encodes the algebraic complexity of the polynomial system. Higher rank implies more variables or constraints, necessitating higher-degree SOS relaxations to capture the optimal solution's structure. This links geometric complexity to algorithmic efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fae63ec2.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fae63ec2.py", line 48, in run_trial
    M_rank = rank(A)
             ^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fae63ec2.py", line 32, in rank
    A = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fae63ec2.py", line 22, in gaussian_elim
    A[i], A[max_row] = A[max_row], A[i]
                       ~^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError during Gaussian elimination, preventing data collection | next: Debug gaussian_elimination function to handle empty/max_row edge cases

---

Free Entropy Gap in Read-Twice Branching Programs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-09 08:02 UTC
- **Entry ID**: `df8fcd796bb4`

Statement

For a read-twice BP P over n variables, define ρ(P) as the free entropy of its transition matrix's spectral measure. Then ρ(P) ≤ log n + O(1), but for the IP_2 BP (inner product mod 2), ρ(P) ≥ n/2 - O(log n).

Rationale

Free entropy quantifies non-commutative independence, which may distinguish read-twice BPs (with structured dependencies) from the highly entangled IP_2 BP. The spectral measure's free entropy could expose hidden correlations in BP transitions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2d7aba2.py", line 89, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2d7aba2.py", line 75, in run_trial
    free_ent = free_entropy(A)
               ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2d7aba2.py", line 54, in free_entropy
    max_row = i + A[i:].index(max(abs(row[i]) for row in A[i:]))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ValueError: 1 is not in list
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples | next: Fix the free_entropy implementation to handle edge cases in matrix row processing

---

Plethysm Coefficient Gap in Symmetric Squares of Permutation Representations for Random CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Plethysm Theory × Boolean Circuit Complexity
- **Recorded**: 2026-05-09 08:36 UTC
- **Entry ID**: `661b05479d3e`

Statement

For random CNF formulas with m clauses and n variables, let ρ(f) denote the plethysm coefficient of the symmetric square of the permutation representation of f's clause incidence matrix. Then ρ(f) ≥ Ω(n^{1.5}) for satisfiable formulas, while ρ(f) ≤ O(n^{0.5}) for unsatisfiable formulas.

Rationale

Plethysm coefficients capture symmetric power decompositions of permutation representations, which may reveal structural differences between satisfiable and unsatisfiable CNFs. The symmetric square's decomposition could expose algebraic complexity gaps tied to circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_988224e6.py", line 124, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_988224e6.py", line 84, in run_trial
    clauses = [generate_clause() for _ in range(m)]
               ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_988224e6.py", line 80, in generate_clau
    if variables not in clause_set and variables[::-1] not in clause_set:
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: unhashable type: 'list'
```

Judge reasoning

Test crashed due to unhashable list type, preventing data collection | next: Fix list hashing issue in generate_clause() by using tuples instead of lists

---

Fourier Min-Coefficient Inverse Proportionality to CNF Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis on Boolean Functions × Boolean Circuit Complexity
- **Recorded**: 2026-05-09 08:50 UTC
- **Entry ID**: `15c4b78718ca`

Statement

For every CNF formula Φ with n variables and m clauses, the minimal absolute Fourier coefficient μ(Φ) satisfies μ(Φ) ≥ c * m^{-1/2} for some absolute constant c > 0. This bound is tight up to polylog(m) factors.

Rationale

Fourier coefficients capture correlation with parity functions, while CNF size reflects combinatorial complexity. The inverse square-root relationship suggests structural constraints on how functions can be represented with sparse clauses.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `241.58s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

---

Kronecker Coefficient Exponential Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Complexity
- **Recorded**: 2026-05-09 09:25 UTC
- **Entry ID**: `f07a47a73b4f`

Statement

For all n ≥ 2, the Kronecker coefficient g(λ, μ, ν) for partitions λ ⊢ n^2, μ ⊢ n, ν ⊢ n of the symmetric power S$^{n(S}$n(ℂ^2)) decomposed into irreducibles satisfies g(perm_n, μ, ν) ≥ 2^{Ω(n)} while g(det_m, μ, ν) ≤ 2^{O(√n)} for m < n^{1.5}.

Rationale

The permanent's symmetric power decomposition exhibits exponentially larger Kronecker coefficients due to its non-GL(n)-invariant structure, while the determinant's decomposition stabilizes under symmetric powers. This reflects inherent complexity differences in their algebraic representations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-9`, elapsed: `183.16s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing verification of support fraction or counterexamples. | next: Re-run test with increased resource limits and debug logging to identify crash cause

---

Algebraic Connectivity Exponentiates DPLL Tree Size for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × DPLL Search Tree
- **Recorded**: 2026-05-09 09:46 UTC
- **Entry ID**: `2fa20a1ca794`

Statement

For a Tse, the DPLL search tree size is at least 2^{c * μ(G)} where μ(G) is the algebraic connectivity of G and c > 0 is a constant.

Rationale

Algebraic connectivity quantifies graph expansion, which is known to correlate with resolution complexity. Higher expansion implies longer resolution proofs, as seen in expander graphs requiring exponential refutations. Thus, the DPLL tree size should reflect this exponential dependence.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.13s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d23d2ec.py", line 124, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d23d2ec.py", line 104, in run_trial
    clauses = generate_tseitin_formula(n)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d23d2ec.py", line 67, in generate_tsei
    clauses.append([-literals[i], -literals[j]])
                    ^^^^^^^^^^^^
TypeError: bad operand type for unary -: 'str'
```

Judge reasoning

Test crashed due to TypeError in generate_tseitin_formula, preventing data collection | next: Fix the string-to-integer conversion error in generate_tseitin_formula and rerun tests

---

Submodular Width of Polymatroid Rank Function and Monotone Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polymatroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-09 09:58 UTC
- **Entry ID**: `9efd21df5cc8`

Statement

For a k-CLIQUE CNF instance Φ, the submodular width of its associated polymatroid P_Φ is Ω(n), while for general CNFs, it is O(log n).

Rationale

The submodular width of P_Φ captures the combinatorial structure of Φ that makes it resistant to efficient monotone circuit computation, mirroring the inherent complexity of k-CLIQUE.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `241.53s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and debug crash cause

---

Symmetric Group Fourier Coefficient Sum Distinguishes Read-Twice from Read-Once BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Read-Twice Branching Programs
- **Recorded**: 2026-05-09 10:06 UTC
- **Entry ID**: `399f95beac0c`

Statement

For a read-twice BP P over n variables, the sum of absolute values of its Fourier coefficients in the symmetric group representation is Ω(2^{n/2}), whereas for read-once BPs it is O(n^2).

Rationale

The symmetric group's representation theory captures permutation symmetries inherent in read-twice BPs, which are absent in read-once. Fourier coefficients encode these symmetries, creating a gap exploitable for separation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_362a060f.py", line 143, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_362a060f.py", line 119, in run_trial
    ft_read_twice = fourier_transform(read_twice_bp, n)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_362a060f.py", line 108, in fourier_tran
```

```
    representations = symmetric_group_representation(n)
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_362a060f.py", line 93, in symmetric_gro
    representation[apply_permutation(cycle, [i])[0]][i] = sgn(cycle)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_362a060f.py", line 86, in apply_permuta
    return [bp[permutation[i]] for i in range(len(bp))]
            ~~^^^^^^^^^^^^^^^^
IndexError: tuple index out of range
```

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity. | next: Debug and fix the symmetric_group_representation implementation to handle permutations correctly

---

Fourier Coefficient Sum Inverse Proportionality to DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Harmonic Analysis on Finite Groups × DPLL Search Tree Size
- **Recorded**: 2026-05-09 10:43 UTC
- **Entry ID**: `2adc13aba337`

Statement

For a random CNF formula Φ with n variables and m clauses, the expected size of the DPLL search tree T(Φ) satisfies E[T(Φ)] = Θ(1 / Σ_{S} |f_S|), where f_S is the Fourier coefficient of the boolean function f corresponding to Φ at subset S.

Rationale

The Fourier coefficients capture the influence of variable subsets on the function's output. A higher sum of absolute coefficients indicates more balanced influence, potentially leading to a smaller DPLL tree as the algorithm can prune branches more effectively.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9bbbe86d.py", line 95, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9bbbe86d.py", line 72, in run_trial
    dpll_size = sum(dpll(cnf, {}) for _ in range(10))
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9bbbe86d.py", line 72, in <genexpr>
    dpll_size = sum(dpll(cnf, {}) for _ in range(10))
                    ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9bbbe86d.py", line 41, in dpll
```

```
    pos_count = sum(sign == 1 for _, sign in clause if literal == abs(literal))
                                             ^^^^^^
NameError: name 'clause' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'clause' in dpll function, preventing data collection | next: Fix the dpll function's clause reference and re-run experiments

---

Positive Eigenvalue Count Inverse Proportional to SOS Refutation Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Refutation Degree for Max-CUT
- **Recorded**: 2026-05-09 11:18 UTC
- **Entry ID**: `de5bbf7852f2`

Statement

For a random Max-CUT instance on n variables, the number of positive eigenvalues of its degree-d moment matrix is ≥ Ω(log n / d^2). Conversely, if an instance's moment matrix has ≤ O(log n / d^2) positive eigenvalues, then its SOS refutation degree exceeds d.

Rationale

The moment matrix's spectral structure encodes polynomial positivity constraints. Positive eigenvalues correspond to feasible directions in the SOS hierarchy, and their count provides a geometric obstruction to low-degree refutations. This bridges algebraic geometry's spectral analysis with SOS complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_94717367.py", line 103, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_94717367.py", line 80, in run_trial
    refutation_degree = sos_refutation_degree(M, d)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_94717367.py", line 64, in sos_refutatio
    A = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_94717367.py", line 31, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError during Gaussian elimination, preventing result evaluation | next: Debug gaussian_elimination() for singular matrix handling

---

Matroid Rank Gap in Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone DNF Formulas
- **Recorded**: 2026-05-09 11:32 UTC
- **Entry ID**: `bdf58ab07b35`

Statement

For any monotone DNF formula with n variables and m clauses, the rank of the matroid formed by its clause incidence matrix is $O(\log n)$ if $m = \mathrm{poly}(n)$, but $\Omega(n)$ for the k-CLIQUE indicator formula with $k \geq 2\sqrt{n}$.

Rationale

Matroid rank captures combinatorial dependencies between clauses. For sparse DNFs, the rank grows logarithmically due to union-closed set properties, but k-CLIQUE requires exponentially many clauses to represent its clique structure, forcing linear rank growth.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_99efd595.py", line 77, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_99efd595.py", line 48, in run_trial
    rank = gaussian_elimination(matrix)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_99efd595.py", line 30, in gaussian_elim
    factor = matrix[j][i] / matrix[i][i]
             ~~~~~~~~~~~~~^~~~~~~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division by zero, preventing data collection to assess support fraction or counterexamples. | next: Debug the Gaussian elimination implementation to handle singular matrices, then re-run with diverse seeds

---

Hilbert Leading Coefficient Inverse Proportional to ABP Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry × Algebraic Branching Programs
- **Recorded**: 2026-05-09 11:59 UTC
- **Entry ID**: `74b831d11ae9`

Statement

For a polynomial f computed by an ABP of size S, the leading coefficient of the Hilbert polynomial of the ideal generated by f's relations is Θ(1/S).

Rationale

ABPs encode polynomials as layered structures, and their size relates to the complexity of the polynomial's ideal. The Hilbert polynomial's leading coefficient captures asymptotic growth of the ideal's dimension, which may inversely correlate with the ABP's efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_022f4c53.py", line 108, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_022f4c53.py", line 92, in run_trial
    leading_coeff = leading_coefficient(hilbert_poly)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_022f4c53.py", line 85, in leading_coeff
    return hilbert_poly[0]
           ~~~~~~~~~~~~^^^
TypeError: 'float' object is not subscriptable
```

Judge reasoning

Test crashed during execution, preventing data collection. The error suggests a type mismatch in handling Hilbert polynomial outputs. | next: Fix the leading_coefficient extraction to handle float outputs, then re-run trials with multiple seeds

---

Resultant Degree Exponential Lower Bound for ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Resultant Theory × ACC^0 Circuit Size
- **Recorded**: 2026-05-09 12:07 UTC
- **Entry ID**: `3e977f2b8d85`

Statement

For any CNF formula with n variables, the degree of the resultant of its clause polynomials is at least 2^{Ω(n)}, implying that any ACC^0 circuit computing this resultant must have size 2^{Ω(n)}.

Rationale

Resultants encode the solvability of systems of equations, and their degree reflects the complexity of the solution space. If ACC^0 circuits cannot compute resultants efficiently, this would imply a separation between ACC^0 and NEXP.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_86806a8c.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_86806a8c.py", line 62, in run_trial
    degree = degree_of_polynomial([resultant_poly])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_86806a8c.py", line 49, in degree_of_pol
    degree = sum(abs(lit) for lit in term)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

Test crashed with TypeError, preventing data collection. The error suggests a fundamental issue with polynomial representation. | next: Debug polynomial construction in test_86806a8c.py to handle resultant_poly as a polynomial object rather than an integer

---

Semialgebraic Dimension Inverse Proportional to SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Refutation Degree for Max-CUT
- **Recorded**: 2026-05-09 12:44 UTC
- **Entry ID**: `88b2482f7682`

Statement

For a random Max-CUT instance on n variables, the semialgebraic dimension of the solution set defined by its degree-d SOS moment matrix satisfies dim_{sa}(M_d) ≤ 2^{-Ω(d/n)} · n^2. Equality holds iff the instance is integrality-gap tight for degree-d SOS.

Rationale

Semialgebraic dimensions capture the intrinsic complexity of polynomial constraint systems. Lower dimensions imply stricter algebraic conditions, requiring higher-degree SOS certificates to approximate Max-CUT, aligning with known integrality-gap thresholds for random instances.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.07s

```
name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
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TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'semialgebraic_dimension', 'metric_value': 0, 'instances_tested': 1, 'conject
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0
```

Judge reasoning

Single instance test with $n \leq 2$ cannot validate exponential scaling claims. Metric saturation at 0 renders the bound vacuous. | next: Test with $n \geq 10$ and varying d/n ratios to observe exponential scaling behavior

---

Polymatroid Rank and SOS Refutation Degree for Monotone k-CLIQUE

- **Verdict**: INCONCLUSIVE
- **Bridge**: Polymatroid Theory × SOS Refutation Degree for CSPs
- **Recorded**: 2026-05-09 13:00 UTC
- **Entry ID**: fa4bcf9e9740

Statement

For any CNF formula Φ on n variables, the SOS refutation degree of Φ is $\Theta(\text{rank}(\Phi))$, where $\text{rank}(\Phi)$ is the polymatroid rank of the hypergraph formed by its clauses. For k-CLIQUE instances, $\text{rank}(\Phi) \geq \Omega(n^{1/2})$ while $\text{rank}(\text{DNF}) \leq O(\log n)$ for any monotone DNF formula.

Rationale

Polymatroid rank captures submodular structure of constraints, which SOS hierarchies exploit via semidefinite programming. Monotone k-CLIQUE's inherent combinatorial rigidity creates a gap between DNF and polymatroid rank, suggesting SOS degree lower bounds via rank growth.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3c708286.py", line 92, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3c708286.py", line 58, in run_trial
    k_clique_rank = hypergraph_rank(k_clique_edges)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3c708286.py", line 43, in hypergraph_ra
    new_edges.add(e)
TypeError: unhashable type: 'list'
```

Judge reasoning

Test crashed due to unhashable type error in hypergraph_rank function, preventing data collection | next: Fix the hypergraph_rank function to handle list edges properly (e.g., convert to tuples) and rerun tests

---

Matroid Rank Submodularity Bounds Monotone DNF Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone DNF Formulas
- **Recorded**: 2026-05-09 13:27 UTC
- **Entry ID**: `5d13e98dc56a`

Statement

For any monotone DNF formula Φ with m clauses over n variables, define μ(Φ) = max_{A ⊆ [n], |A| ≤ k} (rank(Φ_A) - |A|), where Φ_A is the restriction of Φ to A and rank(Φ_A) is the maximum number of disjoint clauses in Φ_A. Then μ(Φ) ≤ log n for all Φ with m = poly(n), and μ(k-CLIQUE) ≥ n^{1/2} for any k ≥ 2.

Rationale

Matroid rank functions are submodular, and their growth rates could expose structural limitations in DNF representations. The k-CLIQUE problem's clause structure inherently requires high rank values due to combinatorial explosion in subset dependencies, creating a natural separation between general DNF and clique-specific formulas.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with stricter resource limits

---

Operator Norm Separation for Read-Twice vs Read-Once BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Operator Theory × Branching Programs
- **Recorded**: 2026-05-09 13:33 UTC
- **Entry ID**: `ce017a87b91d`

Statement

For any read-once branching program P over n variables, the operator norm of its communication matrix M_P is O(log n). For any read-twice branching program Q computing the same function, ||M_Q||_op ≥ Ω(n).

Rationale

Operator norms capture structural differences in communication complexity, with read-once BPs constrained by logarithmic spectral properties while read-twice BPs exhibit linear growth due to repeated state access. This mirrors discrepancy gaps but through spectral analysis.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bfd64f1e.py", line 98, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bfd64f1e.py", line 70, in run_trial
    norm_ro = operator_norm(M_ro)
              ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bfd64f1e.py", line 39, in operator_norm
   U, _, Vt = gaussian_elimination([row + [0] * (len(matrix) - len(row)) for row in matrix])
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bfd64f1e.py", line 31, in gaussian_elim
    A[i][j] /= factor
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division by zero error, preventing result collection | next: Fix numerical stability in Gaussian elimination implementation

---

Convex Body Volume Lower Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometry of Numbers × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 13:46 UTC
- **Entry ID**: `70f3d3ccad21`

Statement

For every n ≥ 1, the communication complexity of DISJ_n is at least log(V(n)), where V(n) is the volume of the convex body defined by {x ∈ [0,1]^n | ∃y ∈ [0,1]^n, x_i + y_i ≤ 1 ∀i} under the ℓ_1-norm. This bound is tight up to polylog(n) factors.

Rationale

The geometry of numbers provides natural measures of 'complexity' for high-dimensional convex bodies. By linking the communication matrix's structure to a convex body's volume, we expose a geometric invariant that could reveal inherent information-theoretic barriers in solving DISJ_n.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `7.81s`

```
'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds': False,
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'
TRIAL: {'metric_name': 'log(V(n))', 'metric_value': None, 'instances_tested': 30, 'conjecture_holds'

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_080a7c92.py", line 83, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_080a7c92.py", line 83, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed due to KeyError before producing valid results; cannot confirm conjecture status. | next: Fix the test's seed handling and re-run with proper error handling for metric computation

---

Free Entropy Lower Bound on Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 14:30 UTC
- **Entry ID**: `50dd985f0904`

Statement

For any $n \times n$ matrix M representing a DISJOINTNESS instance, the free entropy $\phi(M) = -\int \log(\det(M + tI))\, dt$ satisfies $\phi(M) \geq \Omega(\sqrt{n})$ with equality iff M is a rank-1 matrix. The conjecture quantifies the exponential gap between $\phi(M)$ and the communication complexity of M.

Rationale

Free entropy captures spectral properties of random matrices, which may encode the entanglement structure of DISJOINTNESS instances. The integral over $\det(M + tI)$ relates to the matrix's eigenvalue distribution, which could mirror the information-theoretic cost of coordination in communication protocols.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_963dfb76.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_963dfb76.py", line 41, in run_trial
    phi_M = adaptive_quadrature(f, -10, 10) / (2 * math.pi)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_963dfb76.py", line 35, in adaptive_quad
    s = 0.5 * (f(a) + f(b))
               ^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_963dfb76.py", line 30, in f
    M_t = [[M[i][j] + t * delta(i, j) for j in range(n)] for i in range(n)]
                          ^^^^^
NameError: name 'delta' is not defined
```

Judge reasoning

Test crashed due to undefined 'delta' function, preventing evaluation of the conjecture's validity. | next: Define the Kronecker delta function in the code or debug the matrix construction logic

---

Kronecker Coefficient Exponential Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Group × Boolean Circuit Size
- **Recorded**: 2026-05-09 15:27 UTC
- **Entry ID**: `02a5c9227ac1`

Statement

For any CNF formula Φ with n variables, the Kronecker coefficient of the tensor product of the symmetric power S^k(ρ) and S^m(ρ) (where ρ is the standard representation of S_n) is exponentially larger for the permanent than for the determinant when k = n^{0.6} and m = n^{0.4}. This exponential gap implies that the circuit size of Φ is at least 2^{Ω(n)}.

Rationale

Kronecker coefficients capture the multiplicity of irreducible representations in tensor products, which could reflect the structural complexity of the CNF. The exponential gap in these coefficients might indicate that the CNF's structure requires exponentially larger circuits, aligning with the known hardness of the permanent.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
ture_holds': False, 'counterexample': 'Exponential gap not observed'}
TRIAL: {'metric_name': 'Kronecker Coefficient Exponential Gap', 'metric_value': 1.0, 'instances_teste
TRIAL: {'metric_name': 'Kronecker Coefficient Exponential Gap', 'metric_value': 1.0, 'instances_teste
TRIAL: {'metric_name': 'Kronecker Coefficient Exponential Gap', 'metric_value': 1.0, 'instances_teste
TRIAL: {'metric_name': 'Kronecker Coefficient Exponential Gap', 'metric_value': 1.0, 'instances_teste
TRIAL: {'metric_name': 'Kronecker Coefficient Exponential Gap', 'metric_value': 1.0, 'instances_teste
TRIAL: {'metric_name': 'Kronecker Coefficient Exponential Gap', 'metric_value': 1.0, 'instances_teste
TRIAL: {'metric_name': 'Kronecker Coefficient Exponential Gap', 'metric_value': 1.0, 'instances_teste
TRIAL: {'metric_name': 'Kronecker Coefficient Exponential Gap', 'metric_value': 1.0, 'instances_teste

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ec13860c.py", line 75, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ec13860c.py", line 75, in <genexpr>
    first_failing_seed = next(r["see
```

Judge reasoning

Test crashed before producing reliable results, and the single trial's counterexample may not be statistically significant. | next: Run the test with multiple instances and ensure proper seed handling to validate the conjecture

---

Noncommutative Fourier Norm Inverse Proportional to Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Harmonic Analysis × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 16:11 UTC
- **Entry ID**: `250c806b0e11`

Statement

For a |X|×|Y| matrix M defining a DISJOINTNESS instance, let ||F(M)||_op denote the operator norm of its noncommutative Fourier transform over the free group F_2. Then ||F(M)||_op = Θ(1/n) for all n ≥ 2, with constants depending on the group representation parameters.

Rationale

Noncommutative harmonic analysis provides tools to decompose matrix-valued functions on non-abelian groups. By associating DISJOINTNESS matrices with representations of F_2, we can leverage Fourier-analytic bounds to derive communication complexity lower bounds via operator norms.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67f40197.py", line 104, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67f40197.py", line 80, in run_trial
    M = generate_disjointness_matrix(n)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67f40197.py", line 54, in generate_disj
    M[i][j] = 1
    ~~~~^^^
IndexError: list assignment index out of range
```

Judge reasoning

Test crashed due to index error, preventing data collection. Pre-registered support condition cannot be evaluated. | next: Fix the matrix generation code to handle index ranges properly and rerun tests

---

Moment Matrix Sparsity Lower Bound for Max-CUT SOS Approximation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Refutation Degree for Max-CUT
- **Recorded**: 2026-05-09 16:25 UTC
- **Entry ID**: `f58c235db097`

Statement

For any Max-CUT instance with n variables, any degree-d pseudoexpectation satisfying the SOS feasibility conditions must have a moment matrix with $\geq \Omega(n^2)$ non-zero entries. The sparsity threshold is $\Theta(n^2/d^2)$ for $d \geq 2$.

Rationale

The moment matrix encodes polynomial constraints of the SOS hierarchy. High sparsity would imply algebraic dependencies that could be exploited for lower-degree refutations, but the conjecture posits that strong approximation guarantees (like 0.879-approximation) require dense moment matrices, exposing a trade-off between sparsity and approximation quality.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ad841dd.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ad841dd.py", line 65, in run_trial
    Q = gram_schmidt(monomials)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ad841dd.py", line 28, in gram_schmidt
    Q.append([q_i[k] / norm for k in range(len(q_i))])
              ~~~~~~~^~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division by zero error, preventing data collection. Cannot assess conjecture validity without successful runs. | next: Debug gram_schmidt implementation to avoid division by zero, then re-run experiments with multiple seeds

---

Hypergraph Treewidth Inverse Proportional to DPLL Tree Size for Random 3-CNFs

- **Verdict**: INCONCLUSIVE
- **Bridge**: Hypergraph Theory × DPLL Search Tree
- **Recorded**: 2026-05-09 16:43 UTC
- **Entry ID**: 923c457c4dd2

Statement

For a random 3-CNF formula Φ with n variables, the treewidth of its hypergraph representation is inversely proportional to the DPLL tree size for Φ. Specifically, E[treewidth(Φ)] = Θ(1 / E[DPLL_tree_size(Φ)])

Rationale

Hypergraph treewidth captures the structural complexity of the formula, which might influence the branching factor in DPLL search. Lower treewidth suggests more structured clauses, leading to smaller search trees.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.06s

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing data; support fraction cannot be evaluated | next: Run test with extended timeout and higher resource allocation

---

Smallest Singular Value Inverse Proportional to SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Degree for Max-CUT
- **Recorded**: 2026-05-09 16:48 UTC
- **Entry ID**: `b35e8c0854e4`

Statement

For a random graph G(n, p) with adjacency matrix A, the smallest singular value σ_min(A) satisfies σ_min(A) ≥ c * (d)^{-1}, where d is the SOS degree required to approximate Max-CUT with ratio 0.878 ± ε for n ≤ 40.

Rationale

Spectral properties of random matrices influence semidefinite programming relaxations. The inverse relationship between σ_min and SOS degree suggests that graphs with poor spectral concentration require higher-degree SOS proofs to achieve optimal approximations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1ff70dfc.py", line 88, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1ff70dfc.py", line 58, in run_trial
    sigma_min = compute_smallest_singular_value(A)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1ff70dfc.py", line 54, in compute_small
    U, S, Vt = gaussian_elimination(A)
    ^^^^^^^^
ValueError: too many values to unpack (expected 3)
```

Judge reasoning

Test crashed during execution, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the Gaussian elimination implementation to return exactly 3 values and rerun tests with multiple seeds

---

Schur-Weyl Tensor Rank Bounds SOS Degree for Random 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × SOS Refutation Degree for 3-SAT Instances
- **Recorded**: 2026-05-09 17:08 UTC
- **Entry ID**: `d68b5ae25edd`

Statement

For random 3-SAT instances with n variables, the SOS refutation degree d satisfies d ≤ log_2(rank(T)) + O(1), where T is the tensor representation of the clause hypergraph decomposed via Schur-Weyl duality. The rank is computed as the minimal number of symmetric/alternating tensors needed to express T.

Rationale

Schur-Weyl decompositions reveal hidden symmetry in tensor structures, which may constrain the algebraic complexity of refuting CSPs. The rank captures the minimal decomposition complexity, potentially limiting the SOS hierarchy's required degree via representation-theoretic constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.09s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and larger instance sizes to probe scalability limits

---

Geometric Arrangement Discrepancy Inverse Proportional to Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 17:17 UTC
- **Entry ID**: `830213c96f89`

Statement

For a random geometric arrangement of n pseudolines, the discrepancy of the communication matrix for the disjointness problem is Θ(1 / log n), implying that the communication complexity is Ω(log n).

Rationale

Geometric arrangements often exhibit structured discrepancy patterns, and their communication matrices may have inherent geometric constraints that influence the communication complexity of the disjointness problem.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df982df9.py", line 66, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df982df9.py", line 57, in run_trial
    "counterexample": "" if conjecture_holds else f"Discrepancy {discrepancy} does not match expected {
                            ^^^^^^^^^^^^^^^^
NameError: name 'conjecture_holds' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'conjecture_holds', preventing result evaluation | next: Fix the NameError in test code by properly initializing conjecture_holds variable

---

Fourier Coefficient Sum Submodularity for Monotone DNF

- **Verdict**: INCONCLUSIVE
- **Bridge**: Fourier Analysis on Boolean Functions × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-09 17:37 UTC
- **Entry ID**: d5d05615bf79

Statement

For any monotone DNF formula f on n variables, let μ(f) = ∑_{S ⊆ [n], |S| ≤ log n} |f̂(S)|. Then μ(f ∧ g) ≥ μ(f) + μ(g) − μ(f ∨ g) for all monotone DNF f,g, and μ(k-CLIQUE) = Ω(n).

Rationale

Submodularity of μ under conjunction captures trade-offs in variable interactions, while the log n cutoff balances expressiveness and complexity. The k-CLIQUE lower bound stems from its high sensitivity to dense subgraphs, which amplifies Fourier coefficients at large subsets.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s


```
IAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
```

```
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'μ(f) + μ(g) - μ(f ∨ g)', 'metric_value': 0, 'instances_tested': 1, 'conjecture
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0
```

Judge reasoning

parse_fail | next: NONE

Metric Dispersion Lower Bound for Monotone DNF Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Metric Geometry × Monotone DNF Formulas
- **Recorded**: 2026-05-09 17:48 UTC
- **Entry ID**: `735d4d6f849f`

Statement

For any monotone DNF formula F with m terms, the dispersion μ(F) = max_{i≠j} ||x_i - x_j||_2^2 over term vectors x_i ∈ {0,1}^n satisfies μ(F) ≤ O(log m) for m ≤ poly(n), but μ(k-CLIQUE) ≥ Ω(n^2) for all k ≥ 3.

Rationale

High dispersion in term vectors forces geometric separation, making it infeasible to represent k-CLIQUE with sparse DNF. Metric geometry provides a quantitative lens to bound term interactions via packing arguments.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.95s`

```
'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.733333333333334, 'instances_tested': 30, 'co
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.566666666666666, 'instances_tested': 30, 'co
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.133333333333333, 'instances_tested': 30, 'co
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.0, 'instances_tested': 30, 'conjecture_holds
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.3, 'instances_tested': 30, 'conjecture_holds
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.766666666666666, 'instances_tested': 30, 'co
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.333333333333332, 'instances_tested': 30, 'co
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.933333333333334, 'instances_tested': 30, 'co
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.3, 'instances_tested': 30, 'conjecture_holds
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.4, 'instances_tested': 30, 'conjecture_holds
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.3, 'instances_tested': 30, 'conjecture_holds
TRIAL: {'metric_name': 'dispersion', 'metric_value': 28.266666666666666, 'instances_tested': 30, 'co
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: The test found counterexamples with dispersion values consistently exceeding O(log m) for m ≤ poly(n), directly contradicting the conjecture's upper b | next: Investigate the 'mapping_undefined' counterexample to determine its concrete structure and validate its impact on the dispersion lower bound.

Gowers Uniformity Norm Gap in Read-Twice vs Read-Once BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics (Gowers Uniformity Norms) × Read-Twice Branching Programs
- **Recorded**: 2026-05-09 18:13 UTC
- **Entry ID**: `cd21f13f8cbc`

Statement

For any read-twice BP P with size S, the Gowers uniformity norm of order 3 of its characteristic function satisfies ||f_P||*{U^3} = O(log S). For the IP_2 function, ||f_IP2*||{U^3} = Ω(n).

Rationale

Gowers norms capture higher-order correlations critical for functions like IP_2, which require exponential size for read-once BPs. Read-twice BPs, with their limited memory, exhibit pseudorandomness reflected in bounded Gowers norms.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.13s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_83aae375.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_83aae375.py", line 58, in run_trial
    u3_norm_bp = gowers_u3_norm(f_bp, n)
                 ^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_83aae375.py", line 27, in gowers_u3_nor
    F[i][j] += f[(i, j, k)] * math.exp(-2j * math.pi * (i * k + j * k) / n)
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

TypeError: must be real number, not complex
```

Judge reasoning

Test crashed due to type error; no data produced to evaluate conjecture | next: Fix complex number handling in gowers_u3_norm implementation

---

Littlewood-Richardson Coefficient Asymmetry in Permanent vs Determinant Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Symmetric Algebra of Matrices
- **Recorded**: 2026-05-09 18:20 UTC
- **Entry ID**: `95c0d42afb3f`

Statement

For any n ≥ 2, the Littlewood-Richardson coefficient c_λ^{μ,ν} for the decomposition of the permanent's symmetric power S^n(Mat_d(C)) is Ω($2^{\{n}$2}) larger than for the determinant's decomposition when μ = (n^2) and ν = (n^2 - k) for k = O(n).

Rationale

The permanent's symmetric power decomposition exhibits exponentially more symmetries under GL_d(C) than the determinant's, as evidenced by the exponential growth of Littlewood-Richardson coefficients for hook-shaped partitions. This asymmetry could expose algebraic structure exploitable for complexity lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
586, 1585, 1584, 1583, 1582, 1581, 1580, 1579, 1578, 1577, 1576, 1575, 1574, 1573, 1572, 1571, 1570, 15
TRIAL: {'metric_name': 'Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_holds': Fals
TRIAL: {'metric_name': 'Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_holds': Fals
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: All tested instances show ratio=1.0, contradicting the conjecture's claim of Ω(2^{n²}) asymmetry. | next: Verify if perm_coeff and det_coeff are actually equal in these cases, or if there's a miscalculation in coefficient computation

---

Genus Inverse Proportionality to Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry of Curves × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 18:32 UTC
- **Entry ID**: `06752711f237`

Statement

For any n×n communication matrix M of the DISJOINTNESS problem, the genus g(M) of the curve defined by its entries satisfies g(M) ≤ 1/2 $\log_2$ CC(M), where CC(M) is the randomized communication complexity. Equality holds for all complete bipartite graphs.

Rationale

The genus of a curve encodes topological constraints on its defining equations. By linking the genus to communication complexity, we expose geometric obstructions to efficient protocols, as higher genus implies more complex algebraic structure requiring greater communication.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d2e19f50.py", line 115, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d2e19f50.py", line 90, in run_trial
    genus = compute_genus(M)
            ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d2e19f50.py", line 65, in compute_genus
    b = gaussian_elimination(A, b)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d2e19f50.py", line 46, in gaussian_elim
    factor = Augmented[j][i] / Augmented[i][i]
             ~~~~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division-by-zero error, preventing genus computation. Cannot verify conjecture's validity. | next: Debug gaussian_elimination implementation to handle singular matrices, then retest with validated inputs

---

Free Entropy Lower Bound on Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 18:58 UTC
- **Entry ID**: `9020cef6c62f`

Statement

For any $n \geq 1$, the free entropy $\phi(M_n)$ of the DISJOINTNESS matrix $M_n \in \{0,1\}^{n\times n}$ satisfies $\phi(M_n) \geq \Omega(\log n)$. This bound is tight up to constant factors for random instances.

Rationale

Free entropy captures the 'non-commutative complexity' of random variables. By analyzing the DISJOINTNESS matrix's entries as non-commutative random variables, we may uncover structural constraints that mirror communication complexity lower bounds. The logarithmic scaling aligns with known $\Omega(n)$ lower bounds via Forster's theorem.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3542cd42.py", line 79, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3542cd42.py", line 61, in run_trial
    phi_M_n = free_entropy(M_n, n)
              ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3542cd42.py", line 53, in free_entropy
    integral += log_det_tM(t) * dt / t
                ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3542cd42.py", line 48, in log_det_tM
    return math.log(determinant(gaussian_elimination(I_plus_tM)))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ValueError: math domain error
```

Judge reasoning

Test crashed with math domain error, preventing result verification | next: Debug numerical stability in free_entropy computation

---

Symmetric Group Irreducible Component Exponential Gap in Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Monotone Circuit Size
- **Recorded**: 2026-05-09 19:24 UTC
- **Entry ID**: `275156ccf299`

Statement

For n ≥ 4, the number of distinct irreducible representations in the decomposition of the permanent polynomial under S_n is Ω(2^n), while the determinant decomposes into exactly one irreducible representation. This exponential gap persists under linear substitutions.

Rationale

The symmetric group's representation theory reveals structural asymmetry between permanent and determinant. The permanent's decomposition into irreducible components reflects its inherent complexity, while the determinant's single-component structure hints at algebraic simplicity. This asymmetry could translate to computational hardness in monotone circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.53s

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing validation of the conjecture's support fraction or counterexample detection. | next: Increase timeout duration and re-run the test with stricter resource limits to ensure completion.

Quadratic Form Rank Inverse Proportional to SOS Refutation Degree for GF(2) Systems

- **Verdict**: INCONCLUSIVE
- **Bridge**: Quadratic Forms over Finite Fields × SOS Refutation Degree for CSPs
- **Recorded**: 2026-05-09 19:47 UTC
- **Entry ID**: ed5dcc9b878a

Statement

For a random system of m quadratic equations over GF(2), the SOS refutation degree required to prove unsatisfiability is $\Theta(\sqrt{m})$. Specifically, for all $m \geq 2$, the expected refutation degree $E[\delta]$ satisfies $E[\delta] = \Theta(\sqrt{m})$ with constant factors dependent on the quadratic form's rank.

Rationale

Quadratic forms over finite fields encode combinatorial structures that may exhibit hidden algebraic symmetries. Their rank and rank-deficiency could influence the SOS hierarchy's ability to capture dependencies, creating a bridge between algebraic geometry and proof complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62e14fca.py", line 83, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62e14fca.py", line 62, in run_trial
    refutation_degree = sos_refutation_degree(poly_system)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62e14fca.py", line 49, in sos_refutatio
    A = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62e14fca.py", line 25, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed during Gaussian elimination with division by zero, preventing data collection. The error suggests flawed implementation of GF(2) arithmetic. | next: Fix Gaussian elimination for GF(2) by replacing division with XOR-based row operations and retest

---

Algebraic Circuit Complexity via Monodromy Action on Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Topology × DPLL Search Tree
- **Recorded**: 2026-05-09 19:56 UTC
- **Entry ID**: `9330b95a0c44`

Statement

For a Tseitin formula derived from a graph G with n vertices, the DPLL search tree depth is at least $2^{\Omega(\mu(G))}$, where $\mu(G)$ is the minimal number of generators of the fundamental group $\pi_1(G)$ under the monodromy action on the universal cover.

Rationale

Monodromy actions encode topological complexity via group actions on covering spaces. Linking this to DPLL depth provides a novel geometric measure of proof complexity, as the universal cover's structure governs backtrack resistance in resolution.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ef54ee1.py", line 157, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ef54ee1.py", line 138, in run_trial
    depth = simulate_dpll(G)
            ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ef54ee1.py", line 122, in simulate_dpl
    pos_var = abs(literal)
              ^^^^^^^^^^^^
TypeError: bad operand type for abs(): 'tuple'
```

Judge reasoning

Test crashed due to TypeError, preventing depth measurement. Support condition cannot be evaluated without successful trials. | next: Fix the simulate_dpll function to handle tuple literals properly and re-run experiments

---

Newton Polytope Vertex Count Inverse Proportional to SOS Degree for Max-CUT

- **Verdict**: INCONCLUSIVE
- **Bridge**: Real Algebraic Geometry (Newton Polytopes) × SOS Degree for Max-CUT
- **Recorded**: 2026-05-09 20:05 UTC
- **Entry ID**: 56b5e6eb5fea

Statement

For a random Max-CUT instance with n variables, the number of vertices of the Newton polytope of the corresponding polynomial is inversely proportional to the SOS degree required to approximate the Max-CUT value with a 0.878-approximation ratio.

Rationale

The Newton polytope encodes the monomial structure of the polynomial, which influences the complexity of the SOS relaxation. A higher SOS degree may be needed for more complex polytopes, leading to an inverse relationship.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.04s


```
e': 'vertex_count', 'metric_value': 182, 'instances_tested': 1, 'conjecture_holds': True, 'counterexa
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 102, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 67, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 53, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 12, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 148, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 125, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 14, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 391, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 55, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 323, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 91, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'vertex_count', 'metric_value': 62, 'instances_tested': 1, 'conjecture_holds'
RESULT: SUPPORTED mean=124.73333333333333 std=104.04131657930687 support_fraction=0.933333333333333
```


Judge reasoning

All trials used n=1 instances, making scaling analysis impossible. High standard deviation (104) suggests metric instability or definition bugs. | next: Test with n≥10 variables to observe scaling behavior between vertex_count and SOS degree

---

Real Stable Polynomial Degree Lower Bound for Max-CUT SOS

- **Verdict**: INCONCLUSIVE
- **Bridge**: Real Stable Polynomials × SOS Degree for Max-CUT
- **Recorded**: 2026-05-09 20:11 UTC
- **Entry ID**: bf0f9125257a

Statement

For every 3-CNF instance Φ with n variables, the SOS degree required to approximate max-CUT within 1-ε must satisfy deg_SOS(Φ) ≥ log_2(ω(Φ)) + 2, where ω(Φ) is the size of the largest clique in the conflict graph of Φ.

Rationale

Real stable polynomials encode positivity conditions that mirror SOS certificates. Their root structure imposes constraints on the hierarchy's ability to capture max-CUT approximations, creating a bridge between algebraic stability and proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.05s`


```
e': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexampl
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'deg_SOS', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': Fa
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=0.0
```


Judge reasoning

Test failed to compute deg_SOS metric due to 'clique_number_undefined' errors, rendering results unreliable. Pre-registered support condition (80% seeds) not met. | next: Fix metric definition for clique number calculation and revalidate test setup

---

Young Tableau Count Inverse Proportional to Monotone Circuit Size for 3-CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Combinatorics (Standard Young Tableaux) × Monotone Circuit Size for 3-CNFs
- **Recorded**: 2026-05-09 20:27 UTC
- **Entry ID**: `50913375ea21`

Statement

For any 3-CNF formula with n variables, the number of standard Young tableaux of shape λ (where λ is the partition derived from the formula's clause structure) is Θ(1/size(MC)), where size(MC) is the minimal monotone circuit size required to compute the formula.

Rationale

The number of standard Young tableaux is connected to the structure of permutations and the combinatorics of the formula. If the formula has a high monotone circuit size, the associated partition λ may have a structure that limits the number of tableaux, suggesting a deeper connection between combinatorial structures and circuit complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_beadacff.py", line 88, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_beadacff.py", line 70, in run_trial
    tableau_count = hook_length_formula(partition)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_beadacff.py", line 51, in hook_length_f
    return int(math.prod(table))
               ^^^^^^^^^^^^^^^^
TypeError: can't multiply sequence by non-int of type 'list'
```

Judge reasoning

Test crashed due to TypeError, preventing data collection. Support condition cannot be evaluated without successful runs. | next: Fix the hook_length_formula implementation to handle list operations properly and re-run experiments

---

Real Root Count Inverse Proportional to SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometry of Polynomials × SOS Degree for Max-CUT
- **Recorded**: 2026-05-09 21:07 UTC
- **Entry ID**: `d9f5244dde36`

Statement

For a random Max-CUT instance on n variables, the expected number of real roots of the corresponding polynomial P(x) is Θ(n^{1/d}), where d is the SOS degree required to approximate the maximum cut with a 0.878-approximation.

Rationale

The SOS hierarchy's degree d relates to the precision of the approximation. Polynomials with more real roots may have simpler structures, allowing lower-degree SOS certificates via real-rootedness-based moment matrix rank bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c564256f.py", line 106, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c564256f.py", line 89, in run_trial
    sturm_seq = sturm_sequence(polynomial_coeffs)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c564256f.py", line 63, in sturm_sequenc
   next_coeff = [-seq[-2][i+1] * (len(seq[-1]) - i - 1) / seq[-1][0] for i in range(len(seq[-
2]) - 1)]
                                                                                 ~~~^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during Sturm sequence computation, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the IndexError in sturm_sequence() by adding input validation for polynomial coefficients

---

Betti Number Inverse Proportionality to DPLL Tree Size in 3-SAT

- **Verdict**: INCONCLUSIVE
- **Bridge**: Algebraic Topology (Persistent Homology) × DPLL Search Tree
- **Recorded**: 2026-05-09 21:15 UTC
- **Entry ID**: 10b91e07b63e

Statement

For a random 3-SAT instance with n variables, the first Betti number of its clause graph satisfies E[Betti_1] = Θ(1 / E[DPLL_size]). Specifically, for all instances with clause density ρ ∈ [1.2, 2.5], the product Betti_1 * DPLL_size is bounded between 0.8n and 1.2n.

Rationale

Persistent homology captures topological features of the clause graph, which may correlate with the structure of the search space, influencing the efficiency of DPLL algorithms. Betti numbers quantify holes in the clause graph, potentially reflecting redundancy in the search tree.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_460d818f.py", line 126, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_460d818f.py", line 83, in run_trial
    u, v = abs(var), abs(clause[(i + 1) % len(clause)])
                                 ^
NameError: name 'i' is not defined. Did you mean: 'id'?
```

Judge reasoning

Test crashed due to undefined variable 'i' before producing results | next: Debug the test code to fix the NameError and re-run experiments

---

Schur-Weyl Decomposition Irreducible Component Gap for Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Discrepancy of Communication Matrices
- **Recorded**: 2026-05-09 21:30 UTC
- **Entry ID**: `386190a1e55e`

Statement

For random n×n matrices A with entries ±1, let λ(A) be the number of irreducible components in the Schur-Weyl decomposition of (A⊗A⊗…⊗A) (n times). Then λ(perm_n) ≥ 2^{Ω(n)} while λ(det_n) = O(n^3), with equality holding for all n ≤ 40.

Rationale

The Schur-Weyl decomposition captures symmetry structures inherent to permanent/determinant matrices. Discrepancy bounds for communication matrices of these functions may be tied to the complexity of their representation-theoretic decompositions, as symmetric group actions encode algebraic dependencies.

Novelty

- Judge: `INCONCLUSIVE` over 9 arXiv hits

Top hits consulted: - [1807.04318v2] On the Discrepancy of Random Matrices with Many Columns - [1503.04226v2] Symmetric Hadamard matrices of order 116 and 172 exist - [2403.03513v3] Spectrum of random centrosymmetric matrices; CLT and Circular law - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent - [2307.01912v3] Yay for Determinants!

Empirical Test

- exit code: `0`, elapsed: `12.78s`


```
ping_undefined'}
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
```

```
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
TRIAL: {'metric_name': 'Irreducible Components', 'metric_value': 1.0, 'instances_tested': 6, 'conjec
RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed=11
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: The conjecture is contradicted by multiple trials with identical counterexamples 'mapping_undefined'. | next: Investigate the 'mapping_undefined' counterexample's origin in the Schur-Weyl decomposition computation

---

Cheeger Constant Exponentiates Resolution Length for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Refutation Length
- **Recorded**: 2026-05-09 21:52 UTC
- **Entry ID**: `3fd17de5d8ec`

Statement

For any connected graph G with n vertices, the Resolution refutation length of the Tseitin formula τ(G) is ≥ 2^{Ω(h(G))}, where h(G) is the Cheeger constant of G. For graphs with h(G) ≤ 1/√n, this bound is ≤ 2^{O(1)}.

Rationale

Cheeger constants capture expansion properties critical for Tseitin formula hardness. Expanders (h(G) ≈ 1) force exponential refutation lengths via the cumulative entropy framework, while non-expanders (h(G) ≈ 0) admit polynomial-length proofs due to low expansion.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [1807.02225v2] Cheeger inequalities for graph limits - [2404.04038v1] Refutability as Recursive as Provability - [2103.14334v4] Invariant subspaces of elliptic systems II: spectral theory - [1411.3530v4] Cheeger constants, structural balance, and spectral clustering analysis for signed graphs

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9c27fc76.py", line 140, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9c27fc76.py", line 112, in run_trial
    adj = generate_random_regular_graph(n, k)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9c27fc76.py", line 26, in generate_rand
    neighbors = random.sample(range(i + 1, min(i + k // 2 + 1, n)), k // 2)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 430, in sample
```

```
    raise ValueError("Sample larger than population or is negative")
ValueError: Sample larger than population or is negative
```

Judge reasoning

Test crashed due to invalid sampling; no data produced to evaluate conjecture | next: Fix generate_random_regular_graph to handle k//2 ≤ n-i-1 for all i

---

p-adic Valuation of Solution Count Modulo p Inversely Proportional to DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: p-adic Analysis × DPLL Search Tree Size
- **Recorded**: 2026-05-09 22:03 UTC
- **Entry ID**: `633842e040fa`

Statement

For a 3-CNF formula Φ with n variables, the p-adic valuation of the number of satisfying assignments modulo p is $\Theta(\log n)$ if and only if the DPLL search tree size is Θ(2^{n/2}).

Rationale

p-adic valuations capture arithmetic structure, which may correlate with the exponential behavior of search trees in DPLL. The valuation reflects solution density modulo p, potentially exposing algebraic regularities in clause interactions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_93fdd2dc.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_93fdd2dc.py", line 52, in run_trial
   phi = [[random.choice([-3*i-1, -3*i+1, 3*i-1, 3*i+1]) for _ in range(random.randint(3, 5))] for _ in
                              ^
NameError: name 'i' is not defined. Did you mean: 'id'?
```

Judge reasoning

Test crashed due to undefined variable ‘i’, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Fix the test code to define variable ‘i’ and re-run experiments

---

Slice Rank Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics (Slice Rank) × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-09 22:15 UTC
- **Entry ID**: `ea46e1d39086`

Statement

For the n×n matrix M representing the DISJOINTNESS problem, the slice rank of M is Ω(n), implying that the randomized communication complexity of DISJOINTNESS is Ω(n).

Rationale

Slice rank, a measure from additive combinatorics, captures the minimal decomposition of a tensor into rank-1 components. By linking it to communication complexity, we may uncover structural constraints on the problem's information-theoretic requirements.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3e07a911.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3e07a911.py", line 63, in run_trial
    rank = slice_rank(matrix)
           ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3e07a911.py", line 52, in slice_rank
    u, v = rank_1_decomposition(sub_tensor)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3e07a911.py", line 40, in rank_1_decomp
    if tensor[i][j] != 0:
       ~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError, preventing reliable results. The crash indicates potential issues in tensor indexing logic that need resolution before further analysis. | next: Fix tensor indexing bounds in rank_1_decomposition and re-validate with controlled test parameters

---

Symmetric Tensor Rank Gap in Permanent vs Determinant Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Boolean Circuit Complexity (Permanent vs Determinant)
- **Recorded**: 2026-05-09 22:28 UTC
- **Entry ID**: `8edb074fa2a1`

Statement

For a homogeneous polynomial f of degree n, let μ(f) denote the multiplicity of the irreducible representation indexed by (n-1,1) in the decomposition of Sym^n(f). Then μ(perm_n) ≥ 2^{Ω(n)} while μ(det_m) ≤ 2^{O(√n)} for all m < n^{1.5}.

Rationale

The (n-1,1) partition corresponds to the standard representation of the symmetric group, which captures the alternating nature of the determinant. The permanent's symmetric structure forces higher multiplicities in this decomposition, while the determinant's antisymmetry suppresses it. This gap could expose algebraic structure in circuit complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1bf997b9.py", line 125, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1bf997b9.py", line 79, in run_trial
    det_tensors[m-1][0][m-1] += 1
    ~~~~~~~~~~~^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing results, preventing validation of conjecture | next: Fix the list index out-of-range error in test_1bf997b9.py and re-run with proper parameter bounds

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Standard Young Tableau Count Exponential Gap in Symmetric Power Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Monotone Circuit Size for Permanent
- **Recorded**: 2026-05-09 22:50 UTC
- **Entry ID**: `c89c1e80d225`

Statement

For every n ≥ 2, the number of standard Young tableaux of shape λ in the decomposition of Sym^n(perm_n) exceeds the count for Sym^n(det_n) by a factor of Ω(2^n).

Rationale

Schur-Weyl duality links representation-theoretic decompositions to tensor ranks. The symmetric power decomposition of permanent/determinant reveals structural asymmetry: permanent's decomposition contains exponentially more tableaux due to its symmetric nature, suggesting inherent complexity gaps in monotone circuits.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.26s


```
, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0, "conjecture_holds
TRIAL: {"seed": 421, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 463, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 503, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 547, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 593, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 631, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 677, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 727, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 773, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 821, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 877, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
TRIAL: {"seed": 929, "metric_name": "Ratio of SYT counts", "metric_value": inf, "instances_tested": 0
RESULT: SUPPORTED mean=inf std=nan support_fraction=1.0
```


Judge reasoning

The test reports 'instances_tested': 0 and metric_value: inf, indicating no computations were executed. The 'inf' metric is meaningless without actual data. | next: Implement concrete instance generation and computation for n ≥ 2 to validate the conjecture empirically

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Plethysm Multiplicity Exponential Gap in Monotone Permanent vs Determinant

- **Verdict**: INCONCLUSIVE
- **Bridge**: Plethysm Theory × Monotone Circuit Size
- **Recorded**: 2026-05-09 23:33 UTC
- **Entry ID**: 807f8a158fa7

Statement

For all n ≥ 2, the multiplicity of the irreducible representation λ = (n-1,1) in the plethysm $\mathrm{Sym}^{n(\mathrm{Sym}}2(V))$ decomposed for V = C^n is Ω(2^n), while for the determinant polynomial det_n, this multiplicity is O(1).

Rationale

Plethysm multiplicities encode symmetry constraints of polynomials. The exponential gap in λ=(n-1,1) multiplicities between permanent (which requires symmetric power decompositions) and determinant (which factors via exterior powers) may reflect inherent structural complexity barriers for monotone circuits.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_421796fb.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_421796fb.py", line 54, in run_trial
    plethysm_multiplicity = plethysm_multiplicity(n, (n-1, 1))
                            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_421796fb.py", line 40, in plethysm_mult
    for partition in partitions(n):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_421796fb.py", line 50, in partitions
    if len(p) > 0 and p[-1] > 1:
                      ^^^^^^^^^
TypeError: '>' not supported between instances of 'tuple' and 'int'
```

Judge reasoning

Test crashed due to type error in partition comparison, preventing data collection | next: Fix the partitions function to handle tuple comparisons properly